

Physics 111B: CO₂ Laser

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This research aims to explore the operational mechanisms and efficiencies of continuous-wave carbon dioxide (CO₂) lasers. The focus is on investigating how these lasers are affected by different gas pressures and the relationship between input power, output power, and the resulting wavelength spectrum. The experiment utilizes a mixture of CO₂, Helium (He), and Nitrogen (N₂) gases within an optical cavity. The main objective is to examine the stimulated emission process and achieve population inversion for generating significant infrared radiation. To conduct the experiment, a detailed alignment process is carried out using a Helium-Neon laser. Measurements are then taken under different pressures, specifically 10.6, 12.5, and 15 mmHg, with input power ranging from 199 to 282 Watts. Through quantitative analysis, the study reveals the efficiencies of output power and the spectral line emissions corresponding to the varying pressures and input powers. The observed output powers range from 0.74 to 1.93 Watts. Key findings of the study include the identification of spectral lines and their dependence on input voltages and gas pressures. The results demonstrate a direct correlation between increased input power and spectral line efficiency, particularly at pressures of 18.65 mmHg. The study also considers error analysis, which highlights the impact of systematic uncertainties such as alignment inaccuracies and equipment limitations on the experimental outcomes. Overall, this research confirms the fundamental principles that govern CO₂ lasers. It emphasizes the crucial role of gas composition, pressure, and power input in determining laser efficiency and spectral characteristics.

I. INTRODUCTION

Lasers, which stands for Light Amplification by Stimulated Emission of Radiation, function based on the principles of stimulated emission and population inversion. In a laser system, atoms or molecules within a medium are stimulated to higher energy levels, and upon returning to lower energy levels, they release photons. To maintain continuous laser operation, there must be a greater number of atoms in the excited state compared to those in lower energy states, which is a common state known as population inversion. This is made possible through an external energy source. The process occurs within an optical cavity, comprised of two mirrors, which enhances the light output through repeated stimulated emissions and the outcome is a tightly focused, coherent, and single-colored light beam. This lab delves into gas lasers, specifically focusing on the continuous-wave carbon dioxide (CO₂) laser. This advanced piece of scientific equipment, while resembling the foundational principles of gas lasers involving optical cavities and the excitation of a lasing medium, stands out due to its unique combination of CO₂, Helium (He), and Nitrogen (N₂) gases which enables the generation of significant infrared radiation. The objective of this laboratory experiment is to validate the fundamental connections between wavelength and power, while also confirming the I-V curve's dependence on variations in gas pressure.

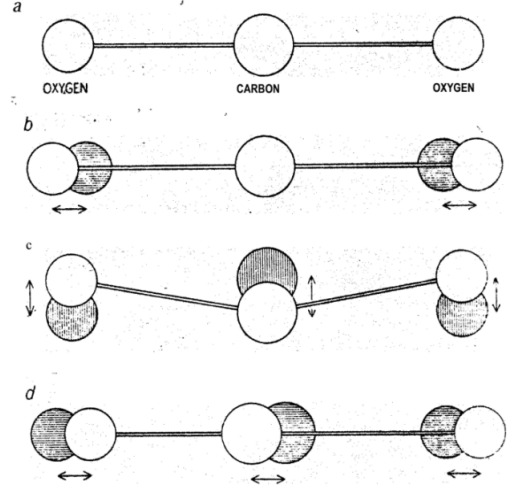


FIG. 1. These are the three vibrational modes of the CO₂ molecule. [1]

II. BACKGROUND AND THEORY

In our experiments with CO₂ lasers, we investigated the mechanisms involved in the changes of vibrational and rotational states in CO₂ molecules, which are crucial for the laser's functionality. The laser's operation depends on the shifts between three vibrational states as seen in Figure 1 - symmetric stretch, bending, and antisymmetric stretch - resulting in the emission of photons at various wavelengths. These different wavelengths

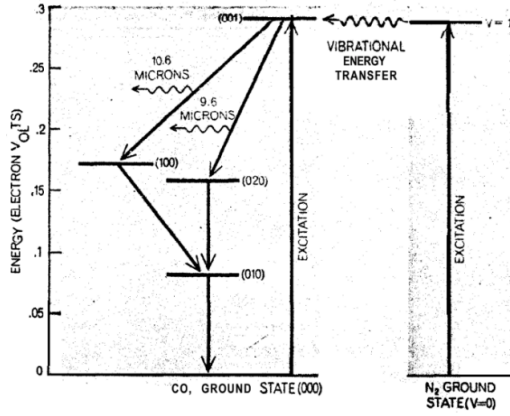


FIG. 2. The energy levels of the CO₂ emissions and their various wavelengths are seen in this image. [1]

lead to different energy emission spectra, and as seen in Figure 2, depending on whether the CO₂ molecule transitions from the $|001\rangle$ state to the $|100\rangle$ or $|020\rangle$ state. Nitrogen molecules play a role in energizing CO₂ from the ground state to the $|001\rangle$ state through collisions, which are facilitated by an external pump and or electrical discharge, while helium assists in removing excess energy by dissipating heat upon colliding with the walls of the tube. Furthermore, the specific restrictions on rotational energy states and transitions of CO₂ also impact the laser's performance, specifically the energy released during transitions. [1]

In this laboratory experiment, we will utilize a diffraction grating as a selective feedback component. This grating enables the reflection of specific wavelengths of light back into the lasing medium for amplification. The selective reflection is achieved by dispersing light into its constituent wavelengths at varying angles, as dictated by the grating equation. By manipulating the angle of the grating in relation to the laser cavity, we can choose the preferred wavelength(s) to be fed back into the medium. While conducting any part of this lab it is important to always keep eyewear goggles on so that your eyes are protected from UV and IR radiation. Furthermore, the CO₂ laser poses a fire hazard, and thus we must ensure there are no stray lasers that can light anything on fire. Furthermore, to ensure safety we have firebricks that we can place in the path of the laser to stop burning, specifically upon powering up the laser, and behind the detectors and mirrors for extra protection. High voltage is another source of extreme danger, so you should always make sure all connections are secure, that the water is flowing, and that gas is flowing through the laser when operating. Make sure you do not touch knob 4 as if you adjust that brass knob you may blow the glass to pieces!

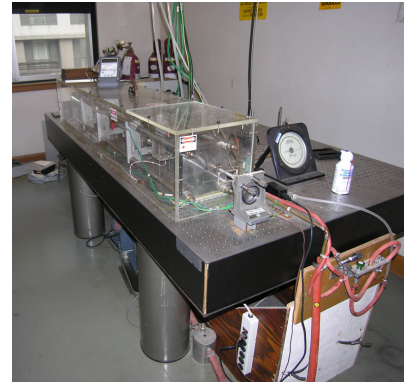


FIG. 3. The full apparatus of the Lab. [2]

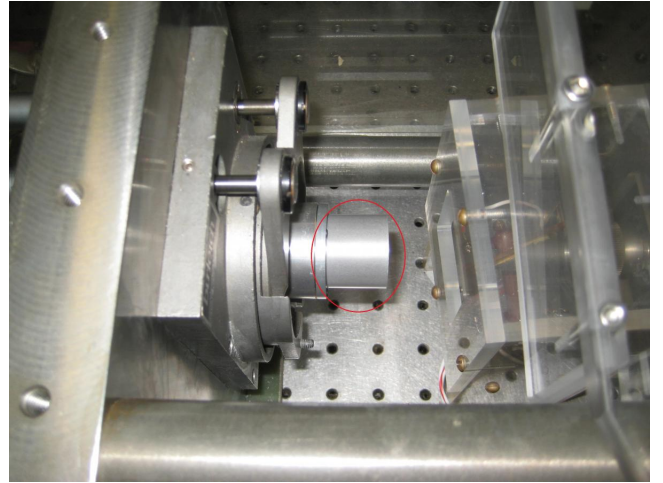


FIG. 4. Picture of the output coupler. [2]

III. EXPERIMENTAL OVERVIEW

A. Setup and Alignment

The whole setup can be seen in Figure 3, and here we see the full apparatus consisting of all the parts used in this experiment besides the high voltage, which we will introduce later when we start lasing.

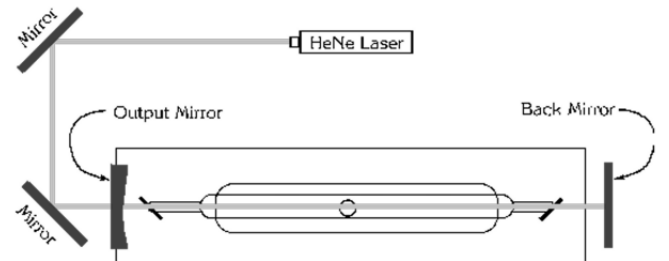


FIG. 5. The top-view of the laser alignment apparatus. [2]



FIG. 6. The Helium-Neon Laser. [2]

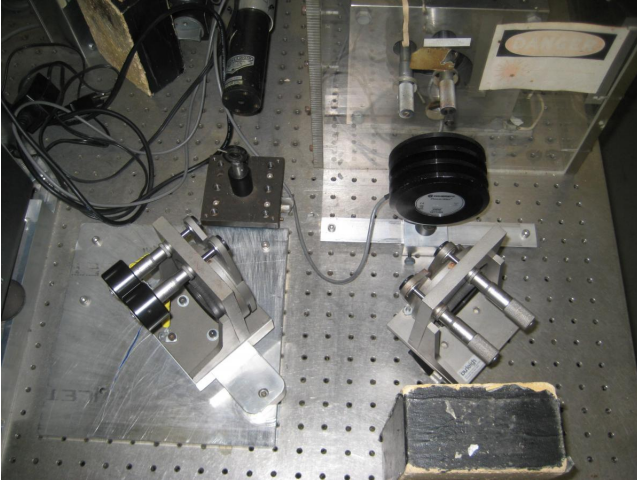


FIG. 7. Optical Alignment Setup. [2]

The first thing we want to do is align the laser, so we first start by opening up the top panels of the encapsulating structure and then we unscrew the output coupler, which is seen in Figure 4. After removing this we want to remove the back mirror or any obstructing object on the opposite end of the tube near the exit area which is seen on Figure 5, so that any laser or light passed through will go straight through and hit the wall. After doing this we then want to go to the Helium Neon laser which can be seen in Figure 6, and use the attached key to turn it on.

Once we have done this we now want to use the optical alignment setup mirrors as seen in Figure 7, and align the right mirror so that the laser will pass through the tube without deflecting off the side, and hit the wall on the other end. Once we have the laser passing through. We want to use the two knobs to either move the laser up and down or left and right. The goal of this is to make the laser an almost perfect dot that passes through and perfectly lines up with the center dot of the penciled circles on the wall as seen in Figure 8. Once we have perfectly



FIG. 8. The laser passing through the tube and hitting the penciled circle on the wall.

aligned this laser, we then want to place the back mirror that we had removed earlier at the end of the setup. We want the He-Ne laser to reflect back through the system and essentially make a halo on the black peephole of the Helium Neon Laser as seen in the top left of Figure 7. We want to use the two knobs of the back laser to align the halo on the peephole, so one partner should look at the peephole, and the other should adjust it until there is a relatively strong halo effect. This halo will look like a red circle surrounding the center peephole, and once you have this you now want to take the output coupler as seen in Figure 5 that you removed earlier and screw it back into the setup. Once this is done you will now see the once halo turn into 2 dots, you want to adjust the output coupler with the knobs that are above it and on the other side so that the bigger and dimmer dot of laser light is perfectly aligned with the peephole. This will complete all the initial alignment prior to the actual lasing.

B. Lasing

Now to begin lasing, there a few prior steps you need to take, first you want to turn on the water in the other room by flipping the lever seen in Figure 9 to be vertical.

After doing this, we now want to pressurize the chamber, so turn the silver knob on the gas tank, so that gas begins to release. This setup can be seen in Figure 10, and the dial hand should quickly increase indicating that gas is releasing into the setup. Now from here, we want to move to the back end of our setup towards the door, where we will find some valves as seen in Figure 11, and



FIG. 9. The main room water line lever. [2]

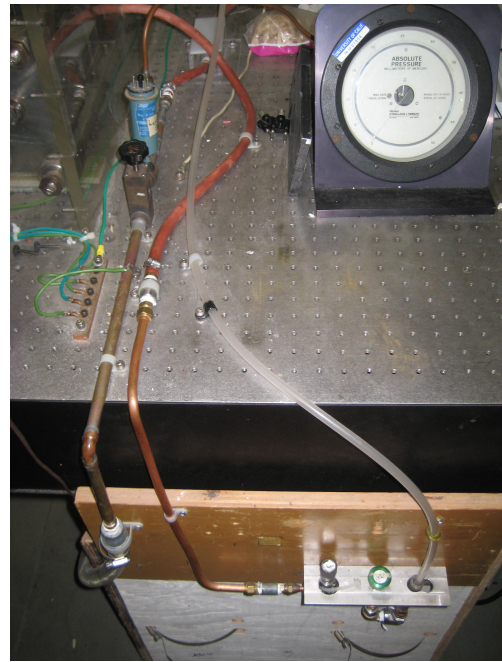


FIG. 11. The gas valves for adjusting pressure. [2]

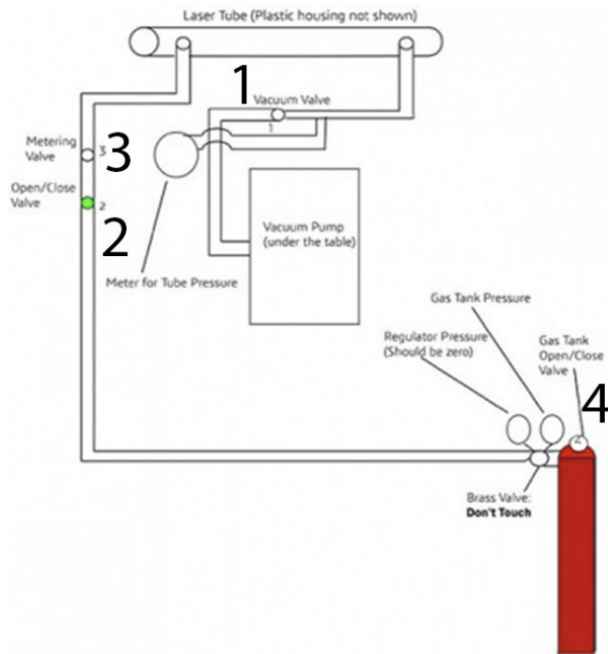


FIG. 10. The gas chamber lines setup. [2]

we will use this to adjust the pressure.

We will use the black knob next to the setup to loosen or tighten, which will hold the pressure in place, decrease or increase it. We will also use the coarse and fine pressure adjustment valves which are the green and silver knobs found at the bottom of Figure 11. First using the black knob we will get the pressure to increase, then we use the green and silver knobs to get the pressure to some number between 10 and 15 mmHg. Once we have ad-



FIG. 12. The high voltage generators setup. [2]

justed the pressure, and the water is on, we now want to use the two switches that are found next to the door and the red button, so while holding the red power button we want to turn the red then the blue lever, and then you should hear water flowing which means that your setup is completely ready.

Now, once this is all ready we are ready to turn on the high voltage as seen in Figure 12, so what we want to do is flip the two regulator switches for the power, then wait about 30 seconds to hear 2 clicks which indicates that they are both on, then turn the High Voltage on the bottom on by clicking the button, and then we can turn



FIG. 13. The power reading of the laser.

the knob until we see the pinkish laser show up in the tube. Once this is done, we are now lasing. We can now move the black power reading scale into the path of the laser and we should be reading a power output, similar to Figure 13. Now you will use the output coupler knobs to fine-tune this power and try to bring it to its maximum possible value, which we found to be around 1-2 W, but it can go as high as 4 W.

C. Measurements

Once you have done this we now are able to take numerous measurements, so by changing the current on the regulators of the high voltage setup, we can log the changes in the power that we read out, and do this for various currents, but we should not go above 10 mA as we found that the fuse will blow if that happens. After we have done this, we now want to record the power output for various pressures. So, to do this we have to first turn off the High Voltage, by flicking the regulator switches off then on, waiting for the voltage to drop down to 0, then flicking them both off, turning the knob to 0 and clicking the generator button so it is all off. Once it is all off, we change the pressure using the valves that we saw in Figure 11, record this and go through the same process as earlier to turn the High Voltage back on, and then record the power output. We repeat this for as many data points as we take. Then we want to analyze the spectrum.

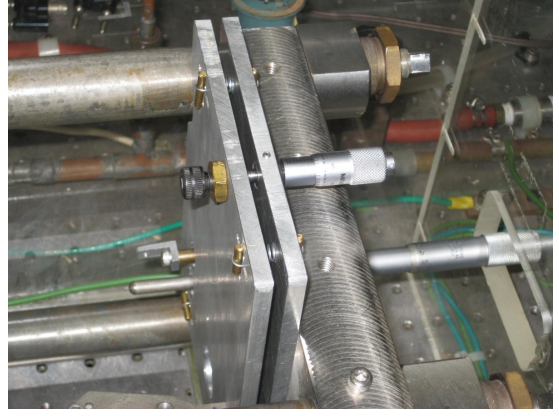


FIG. 14. The diffraction grating. [1]

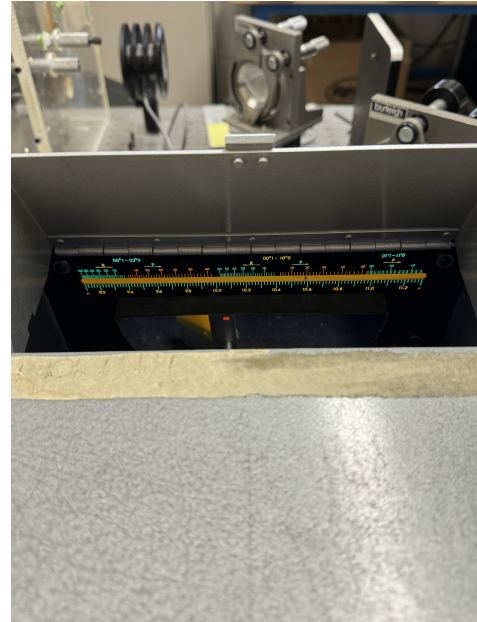


FIG. 15. The diffraction grating.

D. Spectra

Now moving onto this part, we are going to want to turn the High Voltage off like we did earlier, remove the black power readout tool, and the second mirror in the front that directs the laser into the He-Ne, so that the laser is going to bounce off the front mirror and it will bounce into the spectrum analyzer. Now we want to unscrew the end top panel so we have access to the diffraction grating as seen in Figure 14, so we will take some rubber bands and put the grating on and tie it tight so it does not freely move around. Now we will reuse the He-Ne laser to realign, so passing that through the diffraction grating, we will use a paper with a hole in it to let the main laser beam pass through and on the reflected side, we should get a diffraction pattern. Now, we will use the diffraction grating knobs to adjust the laser so that the

first maxima is now being reflected back instead of the central maxima. Once we have overlayed this first maxima laser dot with the entering laser dot, we can now close the panel. Now we want to turn on the spectrum analyzer, use a ruler to make sure it is level, and then we are ready to make detections of spectra. Once this is done, we should be ready to go, so we will now turn on the High Voltage, and begin lasing again, but it is not going to be perfectly aligned into the spectrum analyzer, so we will take the UV light, and we will shine it on the beam finder, which we will then use to detect the exact beam position. Once we have found it, we will keep moving around the beam finder so it is not still for safety purposes, then we will adjust the front mirror using the two knobs so that the laser is perfectly aligned with the hole in the spectrum analyzer, and the laser is now entering the analyzer. Looking at the analyzer, we should begin to see a line form telling us the wavelength of this first maxima, similar to what we see in Figure 15. We then want to go through all of the possible maxima by continuously turning the back knob of the diffraction grating, and as we go through we should see numerous lines begin to form, which is supposed to be representative of the full spectra and wavelengths that we see.

IV. DATA AND ANALYSIS

Collecting data at different pressures, we get the data shown in Figure 1.

Pressure = 10.6		Pressure = 12.5		Pressure = 15	
I (mA)	P (W)	I (mA)	P (W)	I (mA)	P (W)
16	2.02	16	2.18	16	2.14
15	1.78	15	1.93	15	1.9
14	1.74	14	1.8	14	1.68
13	1.55	13	1.6	13	1.41
12	1.3	12	1.39	12	1.34

TABLE I. Current and Power Measurements at Various Pressures

Furthermore, we collect data on the first spectral line and the maxima as shown in Tables 2 and 3. As we varied the input power we noticed a couple of things, specifically, for a pressure of 18.65 mmHg we could not get it to lase even after we set it to 12kV which corresponds to 85 on the knob scale, and on top of that for lower pressures we get it lasing when the knob is at 65 which corresponds to a voltage of 8.7 kV. When comparing this to the output power we noticed that fluctuations were smallest when the pressure was lowest and the power was highest then as well.

Finally, we collect data on the full spectrum by going through the whole diffraction grating as shown in Table 4. When we were working with the spectrum we noticed some things: 1 line is lasing simultaneously, and the lasing spectrum doesn't change when input power is increased. Furthermore, the value changed (decreased)

Input Power (Watts)	Output Power (Watts)
270	1.57
265	1.58
240	1.66
225	1.74
205	1.93

TABLE II. Efficiency at P18 Line

Input Power (Watts)	Output Power (Watts)
199	1.89
225	1.81
243	1.76
259	1.76
282	1.78

TABLE III. Efficiency at P20 Line

when gas pressure decreased, which is what we also expect.

Wavelength (μm)	Output Power (W)
10.21	0.97
10.215	1.02
10.22	1.00
10.23	1.27
10.24	0.96
10.25	0.99
10.26	0.74
10.54	0.56
10.55	0.83
10.56	0.91
10.57	0.86
10.59	1.21
10.6	1.62
10.62	1.32
10.63	1.14
10.64	0.87
10.66	0.79
10.68	0.86
10.69	0.76
10.7	0.78
10.71	0.65

TABLE IV. Output Power vs. Wavelength

Using our data, we craft V-I curves at three different pressure values and use this to determine the relation between the voltage and current at different pressures.

Examining this figure, we see that the chi-squared values tend to be very close to 0, and this is primarily because of our error in the voltage, when discussing with a GSI, we determined that the error in the voltage would be dependent on half of the smallest division that is found in the knob, and since the knob is in kilovolts and there are divisions between each kilovolt, taking half of that we get 50 volts as the error in the voltage, and propagating this through in conjunction with the pressure error which is 0.05 mmHg but is relatively negligible, we see that the chi-squared value is going to be much smaller, and that is why each point also lies within the curve range when

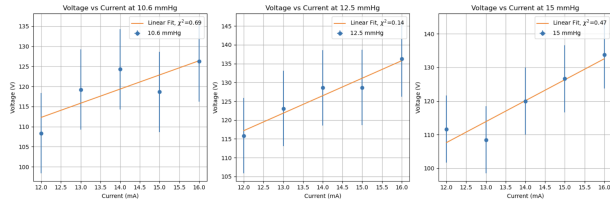


FIG. 16. This is the voltage versus current curves at different pressure values.

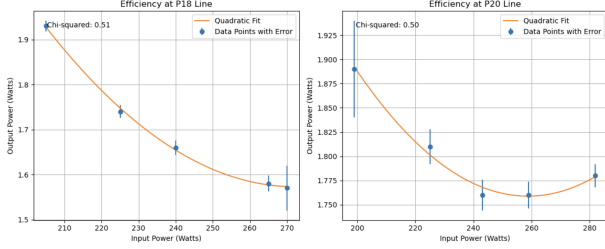


FIG. 17. These are the two spectral lines for the main reflected laser and the first maxima.

accounting for its error bound. But from general examination we can also conclude that these lines are relatively linear in nature which is what we expect it to be as per Ohm's law, so pressure does not have an extremely significant impact on the V-I curves of each of these plots, except for arguably very low pressures where it likely begins to face denaturing, but this is expected for either low or high extremes.

Examining this plot we see that the curves take on a relatively strong quadratic curve, as the chi-squared fits are both around 0.5 which is less than what we hoped for which is 1, but this is to be expected because of the high error associated with the knobs and thus all our points fall within line of what we expect.

Using the collected data we get a plot like the one in Figure 18, and now we can compare this to what we actually expect which can be found in Figure 19, and examining these two plots we see that we get a relatively

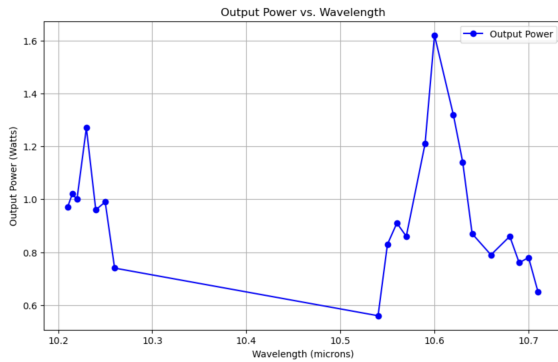


FIG. 18. This is the approximate spectrum we were able to recover from the data.

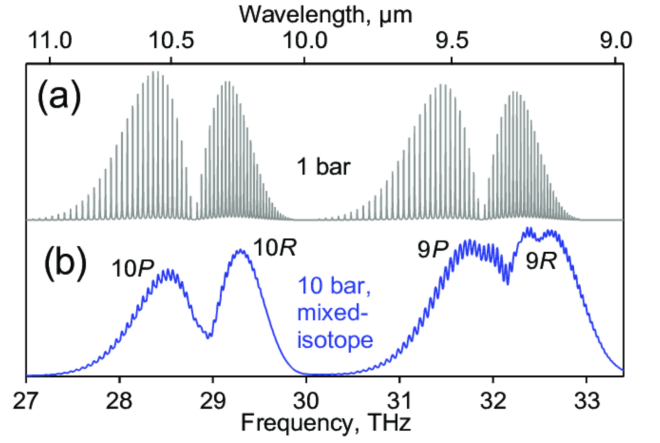


FIG. 19. This the typical gain spectra of CO₂ lasers where image (a) is the conventional atmospheric-pressure laser and image (b) is the high-pressure, mixed-isotope C 16 O₂ - C 16 O 18 O - C 18 O₂ laser. [3]

correct spectrum on the left side of plot a, and it strongly matches what we expect. In fact we spoke to Professor Holzapfel, who mentioned that these plots of the spectrum for the laser look very similar and it is definitely correct for the left half, or higher wavelengths, but we were not able to get the same results for the right side. There can be many reasons for why this is the case which we will discuss in our error analysis.

V. ERROR ANALYSIS

For our error analysis we first need to determine the error in the power. Now our error in power can be propagated through Ohm's Law, but first we need the error in its constituents which is dependent on the knobs. Specifically, the current and voltage: the current is dependent on a knob on the regulator which has marking at each mA, which means our error comes down to the 0.5 mA scale, then our voltage knob has markings ion between each kilovolt so it will be half of 100 volts, which is then 50 volts. Now to calculate the error in power we just need the individual current and voltage and we can determine the error in power.

Given:

$$\Delta V = 50 \text{ volts}$$

$$\Delta I = 0.5 \text{ mA} = 0.5 \times 10^{-3} \text{ A}$$

The error in the power, ΔP , can be found using the formula for the propagation of uncertainty in products:

$$\Delta P = \sqrt{(V\Delta I)^2 + (I\Delta V)^2}$$

Substituting the given values:

$$\Delta P = \sqrt{(V \cdot 0.5 \times 10^{-3})^2 + (I \cdot 50)^2}$$

We use this to get this error formula to get the error in the power measurements we have made, depending on the associated voltage and current from the regulators and generator.

The only other portion that we do not have error for is the pressure, but the pressure has an associated error with the line markings that we set it to, specifically the markings go down to .1 mmHg, so the associated error is 0.05 mmHg. This gives us all the possible error for our calculations, but if our calculations or results are not perfect this can also be attributed to systematic uncertainty in the experiment. One big source of error in this experiment had to do with the setup itself. The laser kept hitting the left side of the outgoing panel which was bolted in place, so we could not do anything to fix that. This causes a slight amount of diffraction off one side, and definitely had an effect on our final results. Furthermore, there were degrees of freedom that were unavoidable, such as the mirror placement, even though it was bolted, there was a slight degree of freedom which can change its placement by up to .5 cm which though it seems small, this can have big impact on laser placement. We also noticed that the spectrum analyzer was not perfectly level which definitely could have had an impact on our spectrum data, and this is not something we could fix

because this was also immovable. There is also gradual fuse burnout issues that would repeatedly happen when we changed the current to too high of a value, and this had an impact on data quality as well (up until the fuse fully burnt out). All of these sources of systematic uncertainty that were more so built into the lab are things that we could not avoid and could have also contributed to the not so perfect results that we see.

VI. CONCLUSION

In conclusion, we can see that the voltage and current curves tend to obey Ohm's law within reasonable error, which is what we expected for the data. Furthermore, we noticed that the efficiency of the spectrum for the p(18) and p(20) lines followed a quadratic curve, as when we increased the input power much higher, the output power decreased less and less, and for the p(20) it eventually even started to increase. Finally, we notice that the spectrum that we found is one half of the actual spectrum. When we were going through the full diffraction grating, at some point we were unable to get any more measurements, and this could be attributed to systematic error, but we were still able to confirm the (CO₂) spectra for the upper half of the wavelength spectra, which is still a very strong result.

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- [1] C. Patel, High-power carbon dioxide lasers, [Scientific American](#) (1968), accessed: 4/8/2024.
 - [2] *CO₂ - CO₂ Laser*, Physics 111B: Advanced Experimentation Laboratory, University of California, Berkeley (2024),

lab manual.

- [3] M. Polyanskiy, I. Pogorelsky, M. Babzien, R. Kupfer, and M. Palmer, Ultrashort-pulse, terawatt, long-wave infrared lasers based on high-pressure co₂ amplifiers, [EPJ Web of Conferences](#) **255**, 11010 (2021).

CO₂ - CO₂ Laser

Signature Sheet

Student's Name Nishank Gite Partner's Name Mathi Raja

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. How does a laser work, in general?
2. What is the purpose of the different gases (i.e. N_2 , CO_2 , and He)?
3. What are the necessary characteristics of the medium, and the cavity?
4. How are the practical requirements for laser action satisfied in a CO₂ laser in this lab?
5. What does a grating do when one is used in place of a mirror in the laser?
6. What are the safety requirements for working with this laser?

Staff Signature VR R Date 03/05/24

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. On day 4 of this lab, you should have successfully aligned the laser and observed a lasing spectrum in the spectrum analyzer. How many lines are lasing? On what does this number depend? Demonstrate your progress to an instructor and ask for a signature.

Staff Signature VR R Date 03/12/24

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Back Mirror Alignment

Staff Signature ✓RR

2. Proper Gas Flow

Staff Signature ✓RR

3. Spectrum Analyzer Signal Line

Staff Signature Alexander

4. P(20) Line

Staff Signature hmt

5. CO₂ Spectrum

Staff Signature Alexander